ways in which we could do it. Firstly, we could ban reproduction before a certain age, say forty. After some centuries of this the minimum age limit would be raised to fifty, and so on. It is conceivable that human longevity could be pushed up to several centuries by this means. I cannot imagine that anyone would seriously want to institute such a policy.

Secondly we could try to 'fool' genes into thinking that the body they are sitting in is younger than it really is. In practice this would mean identifying changes in the internal chemical environment of a body that take place during ageing. Any of these could be the 'cues' that 'turn on' late-acting lethal genes. By simulating the superficial chemical properties of a young body it might be possible to prevent the turning on of late-acting deleterious genes. The interesting point is that chemical signals of old age need not in any normal sense be deleterious in themselves. For instance, suppose that it incidentally happens to be a fact that a substance *S* is more concentrated in the bodies of old individuals than of young individuals. *S* in itself might be quite harmless, perhaps some substance in the food which accumulates in the body over time. But automatically, any gene that just happened to exert a deleterious effect in the presence of *S*, but which otherwise had a good effect, would be positively selected in the gene pool, and would in effect *be* a gene 'for' dying of old age. The cure would simply be to remove *S* from the body.

What is revolutionary about this idea is that *S* itself is only a 'label' for old age. Any doctor who noticed that high concentrations of *S* tended to lead to death, would probably think of *S* as a kind of poison, and would rack his brains to find a direct causal link between *S* and bodily malfunctioning. But in the case of our hypothetical example, he might be wasting his time!

There might also be a substance *Y*, a 'label' for youth in the sense that it was more concentrated in young bodies than in old ones. Once again, genes might be selected that would have good effects in the presence of *Y*, but which would be deleterious in its absence. Without having any way of knowing what *S* or *Y* are—there could be many such substances—we can simply make the general prediction that the more you can simulate or mimic the properties of a young body in an old one, however superficial these properties may seem, the longer should that old body live.